AT100F User Manual V1.2

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Preface

Thank you for choosing the company's AT100F GPS tracking products, please carefully read the instructions before operating.

Please check the items in package with packing list, contact with the distributor when you found something leave out.

⚠ Disclaimer：

- Read this user manual carefully please. When you start use this product, then you are deemed to have read this user's manual.
- This Product used as assistant tool for Security only, can't prevent all kinds of deliberately theft or malicious damage vehicles. For the safety of your assets, you still need keep necessary vigilance and security awareness after you installed this product. We do not bear responsibility to any loss except product itself. Thanks!

1.Product Introduction

AT100F is 4G CAT1 multifunctional GPS tracking device, which combined with positioning, monitor, alarm and tracking functions, with small size it's easy to use and operate, main applies to vehicles and other mobile object location and tracking service.

Product Features:

- Support 9-90V widely power supply.
- Support external power cut alarm.
- Built-in 4G/GPS antenna.
- Support Vehicle mileage statistic.
- Support remote Cut/Recover Engine.
- Support remote upgrade by OTA.
- Support remote configure by SMS/GPRS.
- Bluetooth optional.
- Interfaces:
  ① 2 digital inputs;
  ② 1 digital output.

Note:

- ✓ This product function based on GPRS network; need a SIM card which have GPRS data transmit and SMS functions.
- ✓ SIM card is not include in the packing list, please prepare SIM card before you use this device.
- ✓ This product work voltage is (9-90V/DC), lower or higher will make device work improper.

2.Technical Parameters

Name	Parameters
4G CAT1 SIMCOM	Optional: A7670C LTE-FDD:B1/B3/B5/B8 LTE-TDD:B34/B38/B39/B40/B41 GSM:900/1800 MHz A7670E LTE-FDD: B1/B3/B5/B7/B8/B20 GSM: B3/B8 A7670SA LTE-FDD: B1/B2/B3/B4/B5/B7/B8/B28/B66 GSM: B2/B3/B5/B8
Color	Black
GPS chip	(Zhongke Microelectronics) AT6558R
Working Voltage	9V~90V
Working Current	Around 60mA@12V
Standby current	Around 10mA@12V
Back-up Battery	55mAh
Tracker size	105mm×54mm×20mm
Tracker weight	
Working temperature	-20℃ - +75℃
Moisture	10% - 85%
Locate Sensitivity	5m
Locate time	Hot start(average): <1s Cold start(average): <32s
GPS Sensitivity	-162dbm
GPS frequency	L1:1575.42±1.023MHz
LED Indicator	GSM/GPS/Power
Blind area data	1400/700 records
Bluetooth (optional for Temp&humidity sensor)	• High-performance low-power 32-bit processor • 2.4 GHz transceiver ➢ Compliant to Bluetooth 5.2 ➢ Sensitivity: -97dBm@BLE 1Mbps data rate -103dBm@BLE 125Kbps data rate ➢ TX Power -20 to +10dBm in 3dB steps

	➢ Single-pin antenna: no RF matching or RX/TX switching required ➢ RSSI (1dB resolution) ➢ Antenna array and optional off-chip RF PA/LNA ➢ control interface • Operating temperature: -40 ℃~85℃

3. Define of LED indicator

	LED Color	Status	Status describe
Power status	Blue	Light static	Power Normal
		LED OFF	Power off/in sleep
GSM status	Red	1s Blinking once	Initializing/Finding SIM card
		3s Blinking once	Read SIM card/online
		Fast blinking 0.2s once	Sending GPRS data
		LED OFF	GSM in sleep/GSM fault
GPS status	Green	1s Blinking once	Initializing,finding GPS(not locate)
		3s Blinking once	GPS Located
		LED OFF	GPS in sleep/GSM fault

4. Device Installation

Before you start install the device, Please check the product you get with the packing list, if you have any problem please contact us or distributor in time.

4.1 Prepare works

✧ GPS product AT100F
✧ Local GPRS SIM card
✧ 12V/24V DC power supplier

4.2 SIM Card Installation

This device need install a SIM Card, which must have 4G/GPRS data transmission and SMS function.

① Battery switch;
② SIM slot;
③ USB for configure;
④ USB for download FW.

Note:

- SIM Card must open 4G/GPRS data transmission function
- Confirm the SIM Card have enough balance inside.
- Test SIM card by cellphone to check if it can connect network
- Please do not remove or insert SIM Card without power off, to avoid the damage to SIM card and device

4.3 Device Connection Description

	Color	Describe
1	Red	Power(9-90V)
2	Black	GND
3	Orange	ACC
4	Yellow	OUT1
5	White	SOS
6	Green	AD(0-5V, reserved)

4.4 Device Install Position

To install this device you need have some necessary knowledge about Car Electronics. So please make sure you have right person to make the first installation.
In the installation process, do not power on device. The following is some problem may facing in the installation process, please note:

There have two kind of way to install the device: Hidden install and Open type install. When install in special-purpose vehicle you can select hidden install, and when install in temporary vehicle you can select Open type install.

I. To avoid be broken, the install position of the device should be hidden. The suggest positions are:
   ① Covert within the dalle below the front windshield glass;
   ② Covert around the front instruments panel (the cover of the instruments should not be metal)
   ③ Place Under the dalle below the rear windshield glass of the car.
II. Avoid the positions round emitters, such as reverse sensor, burglar alarm and other vehicle-mounted communication devices.
III. Use the ribbon or sponge powerful double-sided adhesive to fasten the device.
IV. The device GSM Antenna and GPS Antenna are build-in, please make sure the GPS receiving surface (the side with LED indicator) face to sky and no metal shelter above when install.

Note:

✧ If there have metal thermal-protective coating or warm up coating on the windshield glass, the GPS signal will be damped. That may cause the device work abnormal, please change install position.

✧ If you want to install the device by Open type, you can paste the Velcro tape on the dalle below the front windshield glass, and then fasten the device on it.

5.Common Fault analysis

Issue	Analysis	Describe
Can't get online	Power supply abnormal	
	Sim issue	No balance,no GPRS, Install incorrect
	Setting issue	IP,PORT,ID,APN wrong
	GPRS signal weak	
	Server issue	
Can't get location	GPS signal weak	Test with open sky area
	Installation reversed	The antenna side need face to sky

6.Commands

A-Telematics Protocol SMS commands

Name	Command	Reply	Instruct
Set server IP/Port	AS1234*269#1,120.24.225.253,5577,1#	120.24.225.253;5577,1; OK	Reply content is "IP" "Port", connection type: 1: TCP
Set URL/domain	AS1234*269#1,tracking.telematics.net,5577,1#		
Set IP2	AS1234*269#I2,IP,PORT,1,**1**#		
Set Device ID	AS1234*269#2,12007845#	12007845;OK	Will clear history data
Set APN	AS1234*269#3,CMNET,,,#	CMNET,,;OK	APN setting
Restart Device	AS1234*269#B#	GSM/restart; OK	Device restart
Factory set	AS1234*269#F#	Reset OK;	All parameters recover to factory, beside IP/Port/ID/APN
Check connection info	AS1234*269#C#	IP:120.24.225.253,5577,1,APN:M2Minternet,,,0,ID:40677242,M:6,CGR:1,CSQ:18,PPP:1,GPS:A,10,	M6: Online; GPS:A,10: A=locate, V=not locate, 10 located satellite;

		POW:121,N1,30,60,500,40,Z+0.0,AT200-V001	Z+0: Timezone, GMT+0.
Change SMS password	AS1234*269#4,5678#	PWD:5678,OK	
Angle upload	AS1234*269#9,X#	Angle:30, OK	
Position query2	AS1234WHERE2#	http://maps.google.com/maps?hl=en&q=+22.59303,+113.87110	
Open/close Sleep mode	AS1234*269#I,SX#	X=0, close sleep mode; X=1, open sleep mode	Default disable. When open this function, device will turn to sleep mode when ACC off for 5 mins. and wake up when ACC on or alarm or SMS trigger.
Check FW version	AS1234*269#V#	AT200-V001@20231025 IMEI:868768066772420 CCID:89860000192127877768 4G:EC200NCNAAR03A06M08	
Set the authorized SIM number	AS1234*269#5,XXX,XXX,XXX,XXX,#	XXX,XXX,XXX,XXX, set ok	
Authorized SIM number query	AS1234*269#P#	8612345678912,,,,;OK	To check the authorized numbers.
Cut fuel	AS1234ENGINE:OFF#	ENGINE:OFF;OK	Immobilize with duration(5times ON/OFF),and cut oil totally after 30s.
Cut fuel with condition	AS1234*269#O1,1#		
Recover fuel	AS1234ENGINE:ON#	ENGINE:ON;OK	**Recover at once**
Recover fuel with condition	AS1234*269#O1,0#		
Set TIMEZONE	AS1234*269#I,Ex/Wx#	X is 0-12.	Example: set GMT+8 Command: AS1234*269#I,E8#
SMS set ACC ON/OFF upload interval	AS1234*269#7,x,y#	x=ACC ON upload interval, y=ACC OFF upload interval	
Set mileage value	AS1234*269#m,x#	X=mileage value, unit is	AS1234*269#m,1111#

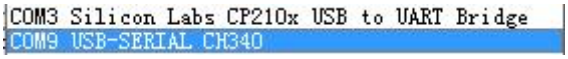
		KM.	MILEAGE:1111(Km);OK
Set mileage ratio	AS1234*269#L,1,X,#	X range: 100-200	AS1234*269#L,1,110,#
Check mileage ratio	AS1234*269#L,0#		MileageRate:110;OK
Set over speed	AS1234*269#OS,x,y,#	X=over speed value; Y=duration trigger time, unit is seconds.	
Set Move alarm(ACC OFF)	AS1234*269#mv,x#	x=move distance, unit is m(ACC OFF, x>100m)	
Set idle alarm	AS1234*269#i,x,y#	x is idle speed, y is duration, unit is minute;	IDLE:8,5;OK
Enable position ACK	AS1234*269#*M,x#	X=1,Open ACK; X=0, close ACK.	
Set harsh accelerate	AS1234*269#a1,X,#	Default: Accelerate:14km/h/s	
Set harsh brake	AS1234*269#a2,X,#	Default: Decelerate:20km/h/s	
GPS harsh corner	AS1234*269#a4,x,y,z#	X=speed, unit is km/h; Y=angle; Z=duration time.	Need update FW.
Query harsh acc/brake parameters	AS1234*269#a#		
OTA CMD	AS1234*269#W,IP,PORT;#		
Check alarm parameters	AS1234*269#R#	D:200m; sp:50,3s; idl:8,3; mv:500; sleep:0; ack:0; acc:0; sos:0; power:0	D:upload by distance; Sp: over speed; idl: idle; Mv: ACC OFF moving alarm distance; Sleep: 1 sleep mode on; Ack: 1 enable position ack; Acc: 1=ACC ON; SOS: 1= SOS ON; Power: 1 main power normal
Mileage unit	AS1234*269#*y,x#	X=0, unit is KM(default); X=1, unit is 0.1km	
Turn over position initialize	AS9876*269#MT,1#	MT:1OK	AS9876*269#MT,1# If turn over for more than 150 and keep for more than 5s will trigger, if keep turn over status will

			keep it till status back.
Disable turn over detect	AS9876*269#MT,0#		
Crash detect	AS9876*269#CA,1,800,1200#	CA:1,X=800 mg,Y=1200 mg Ok	1=enable X:Crash level1, low level crash; Y:Crash level2high level crash; X<Y, suggest X=800mg,Y=X+400mg or more; AS9876*269#CA,1,800,1200# If alarm triggered, 5s
Disable crash detect	AS9876*269#CA,0#		
Query crash param	AS9876*269#CA#		
Binding BT temp&humidity sensor	AS9876*269#BTID,x1,x2,x3,x4,#	[ack] BLE MAC SET OK	MAX support to set 4 sensors
Query BT temp&humidity ID	AS9876*269#BTID#	[ack] BLE MAC:DAF70C,89740A,A2092A,8E8EA1,F067E4, OK	
Clear BTID	AS9876*269#BTID,clear#	[ack] BLE MAC Clear OK	
Serial command format	***269#C**	Remove "**AS1234**" and last "**#**" from SMS command	
Enable log	***269#Y,1**	[com1] len:10 rcv:*269#Y,1 [ack] LOG OK	

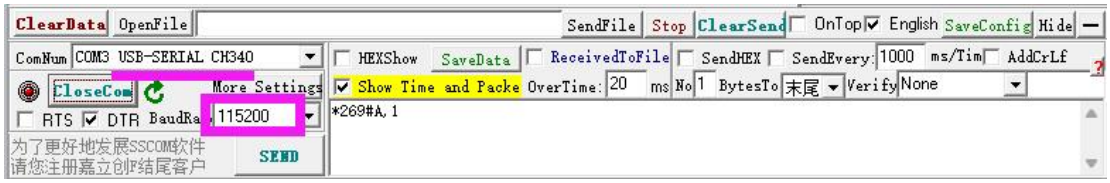
7.Serial Port configuration

- Baud rate:115200;
- Tool: SSCOM or GPS SSCOM;
- USB Driver:

<CH340 USB-SER>



Serial cable:

Enable log with GPS SSCOM:

Enable log with SSCOM command:*269#Y,1